MOMO Quant

Part 6 — UI/UX Specification

Version: 1.0 Draft
Frontend: React + Vite
Styling: Tailwind CSS
Realtime: SignalR
Primary Goal: Build a clear, safe, and usable trading dashboard that helps the developer understand and control the system.

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1. UI/UX Design Principle

The MOMO Quant dashboard must prioritize clarity over decoration.

This is not a marketing website.

This is a control panel for a trading system where bad UI can cause real financial damage later.

The interface must make these things obvious:

- What mode the system is in.
- Whether the bot is running or stopped.
- Whether live trading is disabled or enabled.
- What symbols are active.
- What strategies are active.
- What risk limits are active.
- Why trades are being taken.
- Why trades are being rejected.
- Whether the system is healthy.
- Whether emergency stop is active.

The UI must reduce confusion, not create excitement.

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2. Core UX Rules

The dashboard must follow these rules:

1. Current trading mode must always be visible.
2. Live mode must never look similar to paper mode.
3. Emergency stop must always be accessible.

4. Dangerous actions must require confirmation.
5. Reports must explain performance clearly.
6. Every trade must be traceable to signal, AI decision, risk decision, and order.
7. Every rejected trade must show a reason.
8. Avoid overloaded screens.
9. Use tables for detailed data.
10. Use cards only for summaries.
11. Use charts only where they help decision-making.
12. Do not hide important system warnings.

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3. Visual Identity

Product name:

```
MOMO Quant
```

Recommended tagline:

```
AI-Assisted Quantitative Trading Platform
```

Tone:

- professional
- calm
- technical
- minimal
- serious

Avoid:

- casino-style colors
- excessive animations
- “profit guaranteed” language
- hype-based UI
- flashy trading-bro design

This system should feel like trading infrastructure, not a gambling app.

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4. Layout Structure

Recommended layout:

```
Top Bar
  - Product name
  - Current mode badge
  - Bot status
  - Emergency stop button
  - User profile

Left Sidebar
  - Main navigation

Main Content Area
  - Page content

Right Drawer / Modal
  - Details panels
  - Trade explanation
  - Logs
```

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5. Global Header Requirements

The top header must always show:

```
MOMO Quant
Current Mode
Bot Status
System Health
Emergency Stop
Current User
```

Example:

```
MOMO Quant | PAPER MODE | Bot: Running | Health: Healthy | Emergency Stop
```

Mode badges:

```
Backtest  -> neutral
Replay    -> neutral
Paper     -> safe/demo
Live      -> dangerous/high-visibility
```

Live mode must use a very obvious warning style.

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6. Sidebar Navigation

Initial navigation:

```
Dashboard

Bot Control

Market Watch

Strategies

Backtesting

Replay Mode

Paper Trading

Trades

Orders

Positions

Reports

AI Analytics

Risk Management

Monitoring

Logs

Settings

Admin
```

Admin section includes:

```
Users
Roles
Exchange Settings
API Keys
Audit Logs
```

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7. Dashboard Home Page

The dashboard home page is the control center.

It should show:

7.1 Summary Cards

Cards:

```
Current Mode
Bot Status
System Health
Active Symbols
Active Strategies
Open Positions
Open Orders
Daily PnL
Total PnL
Win Rate
Max Drawdown
Emergency Stop Status
```

7.2 Main Dashboard Sections

Required sections:

```
1. Bot Status
2. Active Trading Session
3. PnL Summary
4. Latest Signals
5. Latest Trades
6. Rejected Trade Reasons
```

```
7. System Health
8. AI Decision Summary
```

The dashboard must not be overloaded with every detail. It should link to detail pages.

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8. Bot Control Page

Purpose:

Control mode, start/stop bot, and trigger emergency stop.

Required controls:

```
Trading Mode Selector
Start Bot
Stop Bot
Pause Bot
Resume Bot
Emergency Stop
Clear Emergency Stop
Active Symbol Selector
Timeframe Selector
Strategy Selector
Risk Profile Selector
```

8.1 Mode Selector

Modes:

```
Backtest
Replay
Paper
Live
```

Rules:

- Backtest, Replay, and Paper can be selected normally.
- Live must be locked until readiness checks pass.
- Live mode requires admin role.
- Live mode requires confirmation text.

• Live mode must show a warning panel.

8.2 Live Mode Warning

Live activation screen must show:

```
You are about to enable LIVE TRADING.
This can place real orders using real money.
Confirm only if you understand the risk.
```

Required confirmation text:

```
ENABLE LIVE TRADING
```

No checkbox-only confirmation. That is too weak.

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9. Market Watch Page

Purpose:

View active market data and symbol status.

Required features:

```
Symbol list
Latest price
24h volume
Spread
Current candle
Timeframe
Market regime
AI confidence
Active strategies
Latest signal
```

Initial symbols:

```
BTCUSDT
ETHUSDT
```

```
SOLUSDT
BNBUSDT
XRPUSDT
```

The symbol list should be configurable.

---

10. Chart View

The chart view should show:

```
Candles
EMA20
EMA50
EMA200
VWAP
Trade entries
Trade exits
Stop loss
Take profit
Signals
Rejected signal markers
```

Chart controls:

```
Symbol
Timeframe
Date range
Strategy overlay toggle
Indicator toggle
Trade marker toggle
Replay mode toggle if applicable
```

Avoid adding too many indicators by default.

The default chart should stay readable.

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11. Strategies Page

Purpose:

Manage strategies and parameters.

Required sections:

```
Strategy list
Enabled/disabled status
Supported regimes
Supported timeframes
Parameter editor
Performance summary
Recent signals
```

Initial strategies:

```
Liquidity Sweep
VWAP Mean Reversion
EMA Pullback
```

Each strategy detail page should show:

```
Description
Entry logic summary
Exit logic summary
Supported regimes
Current parameters
Backtest performance
Paper trading performance
Recent trades
Recent rejected signals
```

---

12. Strategy Parameter Editor

Parameter editor must support:

```
Parameter key
Current value
Value type
Timeframe
Symbol override
Default value
```

```
Last updated
Updated by
```

Changing a parameter must create an audit log.

For risky parameters, show warning before saving.

Examples of risky parameters:

```
MaxRiskPerTradePercent
MinConfidence
AtrMultiplier
OrderTimeoutSeconds
```

---

13. Backtesting Page

Purpose:

Run historical tests safely.

Required inputs:

```
Backtest name
Exchange
Symbols
Timeframe
Higher timeframe
Date range
Initial balance
Strategies
Risk profile
Fee settings
Slippage settings
Maker fill simulation
Order timeout
```

Actions:

```
Run Backtest
Cancel Backtest
View Results
```

```
Duplicate Backtest
Compare Backtests
```

13.1 Backtest Results View

Show:

```
Net PnL
Gross PnL
Fees
Win rate
Profit factor
Expectancy
Max drawdown
Total trades
Average win
Average loss
Largest win
Largest loss
Missed orders
Rejected signals
Performance by symbol
Performance by strategy
Performance by market regime
```

Also show:

```
Equity curve
Drawdown curve
Trade list
Signal list
Rejected reason breakdown
Missed order breakdown
```

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14. Replay Mode Page

Replay mode is one of the most important screens.

Purpose:

Allow the developer to watch historical market data and bot decisions step by step.

Required layout:

```
Chart area
Replay controls
Decision timeline
Current candle details
Current indicators
Current strategy signals
AI decision panel
Risk decision panel
Execution decision panel
Trade log
```

14.1 Replay Controls

Controls:

```
Play
Pause
Step Forward
Step Back if feasible later
Speed: 1x, 2x, 10x, 50x, 100x
Restart
Stop
```

14.2 Replay Decision Timeline

Timeline must show:

```
Time
Market regime
Strategy signal
AI confidence
Risk decision
Execution decision
Trade result
No-trade reason
```

Example:

```
09:33 | BTCUSDT | Trending | EMA Pullback Signal | Confidence 86 | Risk Approved
| Limit Order Submitted
```

Example no-trade:

```
09:36 | BTCUSDT | Choppy | No Trade | Confidence 61 | Rejected: Below confidence
threshold
```

This screen will help debug the bot more than any fancy chart.

---

15. Paper Trading Page

Purpose:

Monitor paper trading like live trading, without real money.

Required sections:

```
Paper account balance
Equity
Open positions
Open orders
Daily PnL
Total PnL
Active symbols
Active strategies
Latest trades
Latest signals
Latest rejected trades
```

Paper trading must clearly display:

```
PAPER MODE — SIMULATED EXECUTION
```

Never let the user confuse paper trading with live trading.

---

16. Trades Page

Purpose:

Review executed trades.

Required table columns:

```
Trade ID
Time
Session
Mode
Symbol
Strategy
Direction
Entry price
Exit price
Quantity
Net PnL
Fees
R multiple
Status
Close reason
```

Filters:

```
Date range
Symbol
Strategy
Mode
Direction
Status
PnL positive/negative
Trading session
```

Trade detail drawer should show:

```
Signal
AI decision
Risk decision
Entry order
Exit order
Fills
```

```
Fees
PnL breakdown
Trade explanation
Chart snapshot if available
```

---

17. Orders Page

Purpose:

Review order lifecycle.

Columns:

```
Order ID
Mode
Symbol
Side
Order type
Price
Quantity
Status
Post-only
Reduce-only
Submitted at
Filled at
Failure reason
```

Order detail should show:

```
Order request
Exchange response if live
Paper/backtest simulation details
Fills
Partial fills
Cancel reason
Missed order relation if applicable
```

---

18. Positions Page

Purpose:

Monitor active and historical positions.

Columns:

```
Symbol
Direction
Quantity
Average entry
Mark price
Unrealized PnL
Realized PnL
Leverage
Margin used
Status
Opened at
Closed at
```

For MVP, leverage can be simulated or fixed. Do not overcomplicate leverage management early.

---

19. Reports Page

Purpose:

Understand whether the bot is actually improving.

Report categories:

```
Performance Report
Strategy Report
Symbol Report
Market Regime Report
Risk Report
Fee Report
Missed Trade Report
Rejected Trade Report
AI Confidence Report
```

19.1 Performance Report

Metrics:

```
Net PnL
Gross PnL
Fees
Funding fees if available
Win rate
Profit factor
Expectancy
Average win
Average loss
Max drawdown
Sharpe ratio
Sortino ratio
Recovery factor
Trade count
Average holding time
```

19.2 Strategy Report

Show performance by strategy:

```
Strategy
Trades
Win rate
Net PnL
Profit factor
Max drawdown
Average R
Best symbol
Worst symbol
Best regime
Worst regime
```

19.3 Rejected Trade Report

Show:

```
Reason
Count
Symbol
Strategy
Timeframe
Estimated missed PnL if measurable later
```

Common reasons:

```
Confidence too low
Risk limit exceeded
Spread too high
Choppy regime
Max open positions reached
Daily loss limit reached
AI anomaly detected
```

This report is essential. Without it, you will not know whether the bot is too strict or not strict enough.

19.4 Missed Trade Report

Show valid signals that failed due to execution logic.

Reasons:

```
Maker order not filled
Price moved away
Order timeout
Spread too wide
Partial fill only
```

This is critical because maker-only scalping can miss many trades.

---

20. AI Analytics Page

Purpose:

Evaluate whether AI is helping or just adding noise.

Required sections:

```
AI Decision List
Confidence Buckets
Market Regime Analysis
AI Approved vs Rejected Outcomes
AI Errors
Parameter Recommendations
Model Versions
```

20.1 Confidence Bucket Report

Buckets:

```
70–79
80–89
90–100
```

Show:

```
Trade count
Win rate
Net PnL
Average R
Max drawdown
False approvals
False rejections
```

If 90–100 confidence trades are not better than 80–89 trades, the AI confidence model is weak and needs improvement.

20.2 Market Regime Report

Show:

```
Regime
Trade count
Win rate
Net PnL
Best strategy
Worst strategy
```

```
Average confidence
Rejected trades
```

This helps confirm whether strategy selection is correct.

---

21. Risk Management Page

Purpose:

Configure and monitor risk.

Required sections:

```
Risk profiles
Risk rules
Current risk state
Daily loss status
Consecutive losses
Open exposure
Correlation exposure
Emergency stop status
```

Risk rules:

```
Max risk per trade
Max daily loss
Max weekly loss
Max open positions
Max exposure per symbol
Max correlated exposure
Max consecutive losses
Minimum confidence score
Maximum spread
Maximum volatility
```

Risk changes must be audited.

---

22. Monitoring Page

Purpose:

Show system health.

Required sections:

```
API health
Database health
Redis health
AI service health
Exchange REST health
Exchange WebSocket health
Worker status
Latency
Error rate
Last market update
Last AI response
Last order update
```

Status values:

```
Healthy
Warning
Critical
Stopped
```

Monitoring must be boring and obvious. Fancy visuals are less important than clear problems.

---

23. Logs Page

Purpose:

Debug system behavior.

Log types:

```
System logs
Trading logs
AI logs
Risk logs
Execution logs
Exchange logs
```

```
Audit logs
Error logs
```

Filters:

```
Date range
Level
Module
Symbol
Trading session
Keyword
```

Important log levels:

```
Info
Warning
Error
Critical
```

---

24. Settings Page

Settings categories:

```
General
Trading
Market Data
Strategies
AI
Risk
Execution
Reporting
Monitoring
Security
```

Settings that affect trading must be clearly marked.

Examples:

```
Default mode
Enabled symbols
```

```
Default timeframe
AI service URL
Order timeout
Maker-only setting
Replay max speed
```

---

25. Admin Pages

Admin pages:

```
Users
Roles
Exchanges
API Keys
Audit Logs
System Settings
```

API key page must never display raw secrets after saving.

Show only:

```
Name
Exchange
Testnet/live
Created at
Last used
Status
```

---

26. Notifications

Notification types:

```
Trade opened
Trade closed
Risk rejection
Emergency stop
System health warning
Exchange disconnected
AI service failed
```

```
Backtest completed
Paper trading started
Paper trading stopped
```

For MVP, in-app notifications are enough.

Email/Telegram/Discord can come later.

---

27. Real-Time UI Requirements

Use SignalR to update:

```
Bot status
Market snapshots
Signals
AI decisions
Risk decisions
Orders
Trades
Positions
Replay ticks
System health
```

The UI must not require manual refresh for active bot monitoring.

---

28. Frontend State Management

Recommended state approach:

```
React Query / TanStack Query for API server state
Zustand or Context for lightweight UI state
SignalR hooks for realtime updates
```

Avoid overengineering with Redux unless absolutely necessary.

For one developer, Redux will likely slow development without much benefit.

---

29. Component Structure

Recommended React structure:

```
src/
  app/
    router.tsx
    providers.tsx

  components/
    common/
    layout/
    charts/
    tables/
    forms/
    modals/
    status/
    trading/
    reports/

  features/
    auth/
    dashboard/
    bot-control/
    market-watch/
    strategies/
    backtesting/
    replay/
    paper-trading/
    trades/
    orders/
    positions/
    reports/
    ai-analytics/
    risk/
    monitoring/
    logs/
    settings/
    admin/

  services/
    api/
    signalr/

  hooks/
```

```
  types/
  utils/
```

Use feature-based organization.

Do not dump everything into generic components.

---

30. UI Component Requirements

Reusable components:

```
ModeBadge
BotStatusBadge
HealthBadge
EmergencyStopButton
MetricCard
DataTable
DateRangePicker
SymbolSelector
StrategySelector
RiskProfileSelector
TimeframeSelector
ConfirmationModal
TradeDetailDrawer
SignalDetailDrawer
AiDecisionPanel
RiskDecisionPanel
OrderStatusBadge
PnLBadge
```

---

31. Safety UX Requirements

Dangerous actions:

```
Enable live trading
Change risk profile
Change risk limits
Start live bot
Clear emergency stop
```

```
Add live exchange API key
Manually close live trade
```

These must require confirmation.

Live trading confirmations must require typed text, not just a button click.

---

32. Empty States

Every page must have useful empty states.

Examples:

No backtests:

```
No backtests yet. Import historical candles and run your first backtest.
```

No trades:

```
No trades found for this session.
```

No AI decisions:

```
No AI decisions available. Start a backtest, replay, or paper session.
```

Good empty states reduce confusion during development.

---

33. Error States

Errors must be specific.

Bad:

```
Something went wrong.
```

Good:

```
Unable to load candles for BTCUSDT 3m. The historical data import may be
incomplete.
```

Every API error should show a useful message.

---

34. Loading States

Use clear loading states for:

```
Backtest running
Historical candles importing
Indicators recalculating
Replay loading
Paper trading starting
Reports generating
AI service responding
```

Do not freeze the UI without feedback.

---

35. MVP UI Scope

Build first:

```
Login
Dashboard
Bot Control
Market Watch
Strategies
Backtesting
Replay Mode
Paper Trading
Trades
Orders
Reports
AI Analytics
Risk Management
Monitoring
Settings
Audit Logs
```

Delay:

```
Advanced notification center
Mobile layout
Dark/light theme customization
Multi-exchange comparison UI
Drag-and-drop dashboard builder
Advanced chart annotation tools
Public user subscription UI
```

---

36. UI Implementation Order

Build UI in this order:

```
1. Layout shell
2. Authentication pages
3. Dashboard home
4. Bot control
5. Exchange and symbol settings
6. Strategy pages
7. Risk settings
8. Market data view
9. Backtesting page
10. Backtest results
11. Replay mode
12. Paper trading page
13. Trades/orders/positions
14. Reports
15. AI analytics
16. Monitoring
17. Logs and audit logs
18. Live trading UI later
```

Do not build live trading UI early. It will create false confidence.

---

37. Final UI/UX Summary

The MOMO Quant interface must act like a cockpit, not a casino.

The dashboard should help answer:

```
Is the bot running?
What mode is it in?
Is the system healthy?
What is it trading?
Why did it take this trade?
Why did it reject that trade?
Is AI helping?
Is risk under control?
Is execution realistic?
Is performance improving?
```

The most important UI rule:

```
Make dangerous actions difficult and important information obvious.
```